

VELSK, Russian Federation

WMO: 228670

Lat: 61.0693N

Lon: 42.0695E

Elev: 298

StdP: 14.54

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.6	-18.2	-29.9	1.0	-24.6	-23.5	1.5	-18.2	13.3	23.4	11.7	24.2	0.4	50	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	84.1	67.4	80.3	65.8	76.8	63.4	69.7	80.1	67.5	77.1	65.7	74.1	4.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	96.7	74.6	64.1	90.5	72.1	62.2	84.6	70.6	33.9	79.9	32.2	77.4	30.7	74.1	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	9.9	8.9	-29.1	88.6	8.1	3.5	-34.9	91.1	-39.7	93.2	-44.2	95.2	-50.1	97.8		
(5)			-29.3	72.2	8.0	2.6	-35.1	74.1	-39.8	75.6	-44.3	77.1	-50.2	79.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.4	11.1	13.4	24.2	37.7	50.0	58.7	64.2	58.9	49.4	37.7	25.0	17.2		
	DBStd	20.88	15.46	14.53	9.65	8.69	9.38	8.04	6.96	6.97	6.87	7.65	11.55	14.21		
	HDD50	5797	1206	1023	801	380	121	15	7	92	387	749	1015			
	HDD65	10205	1671	1443	1266	820	473	223	101	213	469	847	1199	1480		
	CDD50	1208	0	0	0	10	120	275	441	283	74	5	0	0		
	CDD65	139	0	0	0	0	6	33	76	24	0	0	0	0		
	CDH74	1391	0	0	0	3	91	358	706	230	3	0	0	0		
	CDH80	350	0	0	0	0	15	93	185	57	0	0	0	0		
	WSAvg	3.6	3.7	3.8	4.1	3.7	3.8	3.1	2.8	2.7	3.3	4.2	4.1	4.2		
	PrecAvg	23.0	1.4	1.1	1.3	1.3	2.0	2.6	2.8	3.0	1.9	2.1	1.9	1.6		
(15) Precipitation	PrecMax	27.0	2.3	2.4	3.1	4.3	3.5	6.5	6.4	5.7	4.5	4.0	3.5	2.4		
	PrecMin	17.8	0.5	0.4	0.2	0.3	0.9	0.5	0.7	0.8	0.8	0.2	0.4	0.9		
	PrecStd	2.7	0.4	0.5	0.7	0.8	0.7	1.3	1.3	0.9	0.9	0.9	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.7	37.4	48.6	70.9	82.0	87.8	90.0	86.5	73.8	61.0	46.6	41.8		
		MCWB	34.1	35.6	38.9	54.8	61.4	68.0	70.6	69.4	63.1	53.1	44.8	39.7		
	2%	DB	34.3	35.1	42.9	63.0	76.1	82.4	84.9	79.7	67.5	54.6	42.3	37.2		
		MCWB	33.0	33.4	36.1	49.5	58.9	65.2	68.2	66.7	59.0	50.3	40.6	35.7		
	5%	DB	33.1	33.6	38.9	56.4	71.3	77.9	81.3	75.3	63.5	51.0	39.9	34.9		
		MCWB	31.9	32.2	34.0	45.2	55.9	62.8	67.2	63.9	56.7	48.2	38.6	33.4		
	10%	DB	30.7	31.6	36.4	51.6	66.9	73.8	77.7	71.3	60.1	48.2	37.4	33.4		
		MCWB	29.6	30.2	33.0	42.7	54.0	61.1	65.5	62.1	55.1	45.8	36.3	32.2		
	0.4%	WB	34.6	36.1	40.3	55.3	64.8	70.0	72.1	64.6	54.7	45.1	39.8			
		MCDB	35.1	37.1	46.3	69.0	75.7	83.0	85.0	83.0	71.2	59.0	46.5	41.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.2	33.7	37.0	50.3	61.3	67.5	70.6	67.8	60.2	51.1	40.8	35.8		
		MCDB	33.9	34.7	40.9	59.8	71.8	78.8	81.0	76.7	65.5	54.2	41.9	37.0		
	5%	WB	32.1	32.2	35.1	47.0	57.9	65.2	68.6	65.3	57.7	48.5	38.6	33.7		
		MCDB	33.2	33.5	38.2	54.8	67.9	74.7	78.2	73.0	62.2	50.8	39.8	34.6		
	10%	WB	29.6	30.2	33.4	43.6	55.3	62.6	66.8	63.2	55.5	45.8	36.4	32.2		
		MCDB	30.6	31.6	35.8	50.8	65.1	71.6	75.8	69.7	59.4	48.1	37.4	33.4		
	5% DB	MDBR	10.9	12.9	15.4	17.5	19.4	18.6	18.5	16.6	13.5	8.2	7.4	8.9		
		MCDBR	8.6	8.6	16.7	26.5	26.4	25.1	23.8	23.0	19.5	13.3	7.4	6.5		
		MCWBR	8.7	8.0	11.4	15.2	12.2	10.4	10.1	10.5	11.8	9.5	7.0	6.4		
		MCWBR	8.7	8.2	12.6	23.9	22.8	21.8	21.1	19.2	16.9	12.6	7.2	6.7		
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.8	7.9	8.9	14.3	11.8	10.1	10.3	10.0	11.1	9.8	6.8	6.8		
	taub		0.254	0.302	0.329	0.369	0.356	0.354	0.390	0.375	0.342	0.321	0.279	0.226		
		taud	2.027	2.067	2.090	2.203	2.433	2.463	2.374	2.395	2.474	2.384	2.134	1.971		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		142	206	245	259	275	278	262	257	247	207	134	98		
		Edh at Noon	13	24	34	37	32	31	34	30	24	18	12	8		
	RadAvg		89	285	684	1148	1571	1647	1658	1185	672	270	79	43		
(45) All-Sky Solar Radiation	RadStd		15	76	113	130	108	155	135	110	80	37	14	8		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		+1.31	N/A	N/A	N/A	N/A	N/A	-400	-455	N/A	N/A
Regional (5 neighbors)		+1.21	N/A	N/A	N/A	N/A	N/A	-351	-421	N/A	N/A

Nomenclature: See separate page